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What's New in DFSMSHsm

Session ID: SSZ09

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Agenda

- **DFSMSHsm V1R10**
 - DFSMSHsm reporting via DFSMSrmm Report Generator
 - EAV Support
 - CDS Backup Improvement
 - Virtual Concurrent Copy
 - Data Set Backup NEWNAME
 - Small Requirements
- **DFSMSHsm Health Checks**
- **DFSMSHsm V1R9**
 - DFSMSdss Cross Memory
 - RP Exit
 - FSR updates
- **DFSMSHsm V1R8 Overview**
- **Development APARs**

DFSMSHsm Reporting

Generate reports of DFSMSHsm functions and inventory using DFSMSrmm Report Generator

- DFSMSrmm Report Generator is an easy-to-use ISPF application
 - Create and customize reports specific to your needs
 - Available without a DFSMSrmm license
 - New option on ISMF panel to create 'Storage Management' reports
 - Sample Reports shipped in SYS1.SAMPLIB

DFSMSHsm reporting based on

- DFSMSHsm Function Statistics Record (FSR)
- DFSMSHsm ABACKUP/ARECOVER Function Statistics Record (WWFSR)
- DFSMSHsm Inventory (control data set) data via DCOLLECT

DFSMSHsm Reporting *(cont)*

Reporting Records

- **ARCFSR** (SYS1.MACLIB)
 - FSR for DFSMSHsm in SMF form
 - Converted inline to ARCFSR2 by ARCGFSRC
- **ARCWWFSR** (SYS1.MACLIB)
 - WWFSR for ABARS in SMF form
 - Converted inline to ARCWWFSR2 by ARCGWFSC
- **ARCUTILP** (SYS1.MACLIB)
 - Modified with keywords to map each structure
 - TYPE=ALL | M | B | C | T
 - ALL for all structures
 - M for Migrate
 - B for Backup
 - C for DASD Capacity Planning
 - T for Tape Capacity Planning
- ARCGFSRC & ARCGWFSC are shipped in SYS1.DGTSLIB

DFSMSHsm Reporting *(cont)*

Report Types

Name	Description
ARCGWFSR	ABARS
ARCGDBCK	DCOLLECT BACKUP
ARCGDDSD	DCOLLECT DASD CAPACITY
ARCGDMIG	DCOLLECT MIGRATION
ARCGDTAP	DCOLLECT TAPE CAPACITY
ARCGFSR2	DFSMSHsm Functions

DFSMSHsm Reporting *(cont)*

Sample Reports

Name	Report Title	Report Type
ARCGAB01	ABARS ABACKUP Statistics	ABARS
ARCGAR01	ABARS ARECOVER Statistics	ABARS
ARCGDB01	DCOLLECT BACKUP Data	DCOLLECT BACKUP
ARCGDD01	DCOLLECT DASD CAPACITY PLANNING	DCOLLECT DASD CAPACITY
ARCGDM01	DCOLLECT MIGRATION Data	DCOLLECT MIGRATION
ARCGDT01	DCOLLECT TAPE CAPACITY PLANNING	DCOLLECT TAPE CAPACITY
ARCGS001	DFSMSHsm Statistics	DCOLLECT TAPE CAP
ARCGS002	Backup Statistics	DFSMSHsm FSR-SMF Records
ARCGS003	Migration Statistics	DFSMSHsm FSR-SMF Records
ARCGS004	Recall Statistics	DFSMSHsm FSR-SMF Records
ARCGS005	Recover Statistics	DFSMSHsm FSR-SMF Records
ARCGS006	Volume Dump Statistics	DFSMSHsm FSR-SMF Records
ARCGS007	Restore from Dump Statistics	DFSMSHsm FSR-SMF Records
ARCGS008	FRBACKUP Statistics	DFSMSHsm FSR-SMF Records
ARCGS009	FRRECOV Statistics	DFSMSHsm FSR-SMF Records
ARCGS010	DFSMSHsm Thrashing	DFSMSHsm FSR-SMF Records

ABACKUP Sample Report

ABARS ABACKUP Statistics

- 2 -

2007/07/29

10:52:02

AG NAME AGGRP2

DATE	TIME	USERID	C FILE DSN	# TAPE	Host
2007170	10313880	HSMATH0	DFHSM.ABARS.C.C01V0002	4	1
2007171	10304527	HSMATH0	DFHSM.ABARS.C.C01V0003	4	1
2007172	10300561	HSMATH0	DFHSM.ABARS.C.C01V0004	4	1

GROUP TOTALS:

12

AG NAME AGGRP3

DATE	TIME	USERID	C FILE DSN	# TAPE	Host
2007171	10313880	HSMATH0	DFHSM.ABARS.C.C01V0003	4	1

GROUP TOTALS:

4

Other fields included in the sample report:

- Number and K-Bytes of Level-0 data sets
- Number and K-Bytes of Migrated data sets
- Number of Accompany data sets
- Total K-Bytes
- CPU Time
- Return Code / Reason Code

Tape Capacity Report

DCOLLECT TAPE CAPACITY PLANNING

- 1 -

2007/07/29

TYPE TAPE B

# FULL TAPE	# PARTIAL	# EMPTY
184	46	16
0	0	0
GROUP TOTALS:		
184	46	16

TYPE TAPE D

# FULL TAPE	# PARTIAL	# EMPTY
1858	0	6
0	0	0
GROUP TOTALS:		
1858	0	6

TYPE TAPE M

# FULL TAPE	# PARTIAL	# EMPTY
103	36	0
0	0	0
0	0	0
GROUP TOTALS:		
103	36	0

# FULL TAPE	# PARTIAL	# EMPTY
00	82	22
2145		

Migration Data Sample Report

DCOLLECT MIGRATION DATA

- 1 -

2007/07/29

MIG DATE	MIG TIME	DSN	1ST SRC VOL
2007158	14250634	HSMATH0.SMS.VBGPS1	SMS001
2007158	14250646	HSMATH0.SMS.VSMALNA	SMS001
2007158	14250661	HSMATH0.SMS.VSMALNB	SMS001
2007158	14250674	HSMATH0.SMS.VSMALNC	SMS001
2007158	14250686	HSMATH0.SMS.VSMALND	SMS001
2007158	14250698	HSMATH0.SMS.VSMALNE	SMS001
2007158	14250709	HSMATH0.SMS.VSMALNF	SMS001
2007158	14250718	HSMATH0.SMS.VSMALNG	SMS001
2007158	14250732	HSMATH0.SMS.VSMALNH	SMS001
2007158	14250742	HSMATH0.SMS.VSMALNI	SMS001
2007158	14250758	HSMATH0.SMS1.PS.TEST0	SMS001
2007158	14250751	HSMATH0.SMS1.PS.TEST1	SMS001
2007158	14250766	HSMATH0.SMS2.PS.TEST2	SMS001
2007158	14250776	HSMATH0.SMS3.PS.TEST3	SMS001
2007158	14250783	HSMATH0.SMS4.PS.TEST4	SMS001

Other fields included in the sample report:

- K-Bytes of Original data set and Migration copy
- Expiration date
- SMS Class names

Backup Sample Report

Statistics for Backup

- 1 -

2007/07/29

11:41:00

DATE	TIME REQ	TIME COMP	DSN	SOURCE	TARGET
2007158	14145274	14145347	HSMATH0.SMS.VBGPS1	SMS001	MIG101
2007158	14171032	14171808	HSMATH0.SMS.VBGPS1	SMS001	
2007158	14144720	14144744	HSMATH0.SMS.VSMALNA	SMS001	MIG101
2007158	14145008	14145023	HSMATH0.SMS.VSMALNA	SMS001	MIG101
2007158	14170596	14170994	HSMATH0.SMS.VSMALNA	SMS001	
2007158	14170970	14170998	HSMATH0.SMS.VSMALNB	SMS001	
2007158	14170989	14171003	HSMATH0.SMS.VSMALNC	SMS001	
2007158	14170995	14171010	HSMATH0.SMS.VSMALND	SMS001	
2007158	14171001	14171016	HSMATH0.SMS.VSMALNE	SMS001	
2007158	14171007	14171023	HSMATH0.SMS.VSMALNF	SMS001	
2007158	14171012	14171030	HSMATH0.SMS.VSMALNG	SMS001	
2007158	14171019	14171036	HSMATH0.SMS.VSMALNH	SMS001	
2007158	14145814	14145827	HSMATH0.SMS.VSMALNI	SMS001	MIG101
2007158	14171026	14171045	HSMATH0.SMS.VSMALNI	SMS001	

Other fields included in the sample report:

- Number of tapes
- Number of K-Bytes read and written
- SMS Class names
- Return Code / Reason Code

FRBACKUP Sample Report

Statistics for FRBACKUP

- 1 -

2007/07/29

12:14:52

CP NAME COPYPL1

DATE	TIME REQ	TIME COMP	SOURCE	TARGET	RC	REAS	HOSTID
2007198	16124594	16124610	SRC001	TAR001	00000000	00000000	1
2007198	16124594	16124621	SRC002	TAR002	00000000	00000000	1
2007198	16124594	16124633	SRC003	TAR003	00000000	00000000	1
2007198	16124861	16124876	SRC001	TAR004	00000000	00000000	1
2007198	16124861	16124886	SRC002	TAR005	00000000	00000000	1
2007198	16124861	16124896	SRC003	TAR006	00000000	00000000	1
2007199	14070930	14071046	SRC003	TAR003	00000000	00000000	1
2007199	14070930	14071094	SRC001	TAR001	00000000	00000000	1
2007199	14070930	14071096	SRC002	TAR002	00000000	00000000	1
2007199	14072157	14072230	SRC001	TAR004	00000000	00000000	1
2007199	14072157	14072230	SRC002	TAR005	00000000	00000000	1
2007199	14072157	14072231	SRC003	TAR006	00000000	00000000	1

Thrashing Sample Report

Migration Age of zero when data set is recalled

DFSMSHsm Thrashing Report

- 1 -

2008/02/04

15:06:18

DSN	AGE	SIZE KB	MC NAME
HSMATH0.SMS.VBGPS1	0000	36830	MCLASS1
HSMATH0.SMS.VSMALNA	0000	159	MCLASS1
HSMATH0.SMS.VSMALNB	0000	159	MCLASS1
HSMATH0.SMS.VSMALNC	0000	159	MCLASS1
HSMATH0.SMS.VSMALND	0000	159	MCLASS1
HSMATH0.SMS.VSMALNE	0000	159	MCLASS1
HSMATH0.SMS.VSMALNF	0000	159	MCLASS1
HSMATH0.SMS.VSMALNG	0000	159	MCLASS1
HSMATH0.SMS.VSMALNH	0000	159	MCLASS1
HSMATH0.SMS.VSMALNI	0000	159	MCLASS1
HSMATH0.SMS1.PS.TEST0	0000	3	MCLASS1
HSMATH0.SMS1.PS.TEST1	0000	3	MCLASS1
HSMATH0.SMS2.PS.TEST2	0000	3	MCLASS1

Other fields included in the sample report:

- Date; Elapsed time
- Target volume
- Return Code / Reason Code

Get it Now!

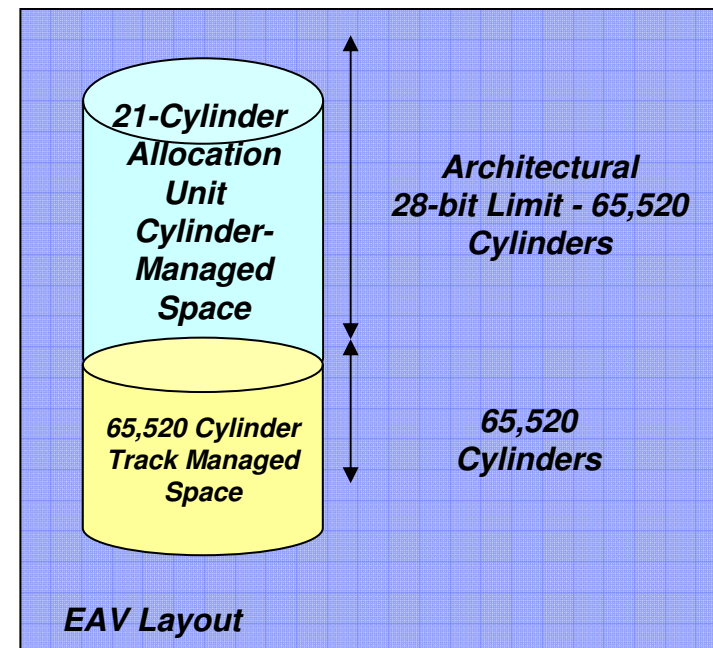
- Copy Libraries from z/OS V1.10 to any current release
- Concatenate in front of existing System Libraries
- Copy Macros to MACLIB to replace existing ones

	DD Name	DDDEF(member name filter)
Panels	ISPPLIB	SEDGPENU(EDG*)
Messages	ISPMLIB	SEDGMENU(EDG*)
Skeletons	ISPSLIB	DGTSLIB(EDG*,ARC*)
Tables	ISPTLIB	DGTTLIB(EDG*)
Execs	SYSEXEC	SEDGEXE1(EDG*)
Load Modules	ISPLLIB	LINKLIB(EDGG*)
Macros		MACLIB(ARCF SR2, ARCWFSR2, ARCUTILP, IDCDO UT, EDGS*)

EAV

Extended Address Volume – A volume for VSAM data sets that tolerates other data set types

- Extended Address Volume
 - No new device type
 - New Model -> 3390 Model A
 - V1R10 Max: 262,668 cyl
- Track-managed space
 - 65,520 cylinders and below
 - All data set types
- Cylinder-managed space
 - Above 65,520 cylinders
 - VSAM data only

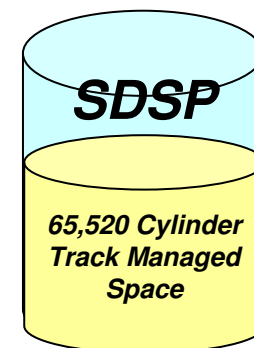
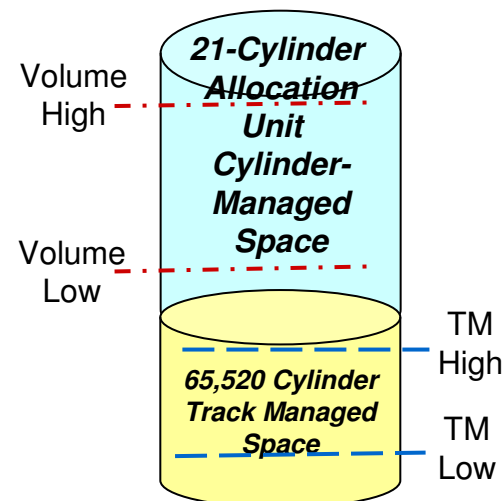


*Track-managed space is same as today.
Cylinder-managed space has 21-cylinder allocation units.*

EAV (cont)

DFSMSHsm Support of EAVs

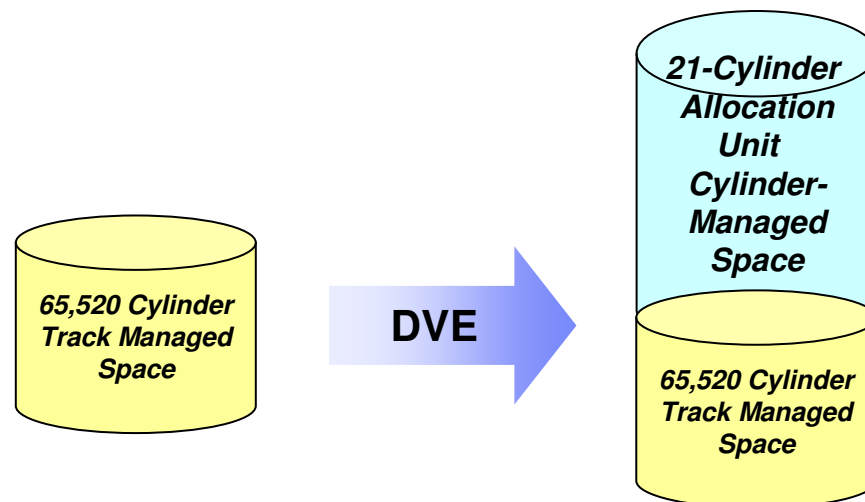
- Space Management: **Two thresholds**
 - Since only VSAM data sets can be allocated in cylinder-managed space
 - High / Low Volume Threshold (existing)
 - High / Low Track-managed Threshold
 - DFSMSHsm will ensure that both the track-managed space *and* entire volume have met their thresholds
 - TRACKMANAGEDTHRESHOLD added to ADDVOL for Primary nonSMS volumes
- SDSPs can be allocated in cylinder-managed space
 - Individual migration copies will only be allocated in track-managed space
 - Don't use EAVs as ML1s until all hosts on V1R10



Dynamic Volume Expansion (DVE)

SPE – OA21574 (November 2007)

- z/OS V1R7 and higher
- IBM DS8000 function that enables a volume to be increased in size *without varying the volume offline*
 - DFSMSHsm listens for an ENF signal that indicates that a volume has increased in size
 - The ENF is triggered by an ICKDSF VTOC REFORMAT
 - After the Reformat, all future DFSMSHsm processing uses the new volume size



CDS Backup Enhancement

- CDS Backup must quiesce all DFSMSHsm activity before it runs
 - Issues an exclusive enqueue
 - Waits for all active DFSMSHsm activity to release shared enqueue
 - Long wait for long-running functions
 - Other DFSMSHsm activity cannot begin after the CDS Backup exclusive enqueue has been issued
- In V1R10, CDS Backup function will use XCF to signal to all DFSMSHsm hosts that CDS Backup is waiting to begin
 - After each data set is processed and at certain intervals, hosts will check if they should temporarily release shared enqueue
 - This enables CDS Backup to begin promptly
 - Existing patches temporarily release enqueues *unnecessarily*
- CDS Backup will still be delayed due to large data sets

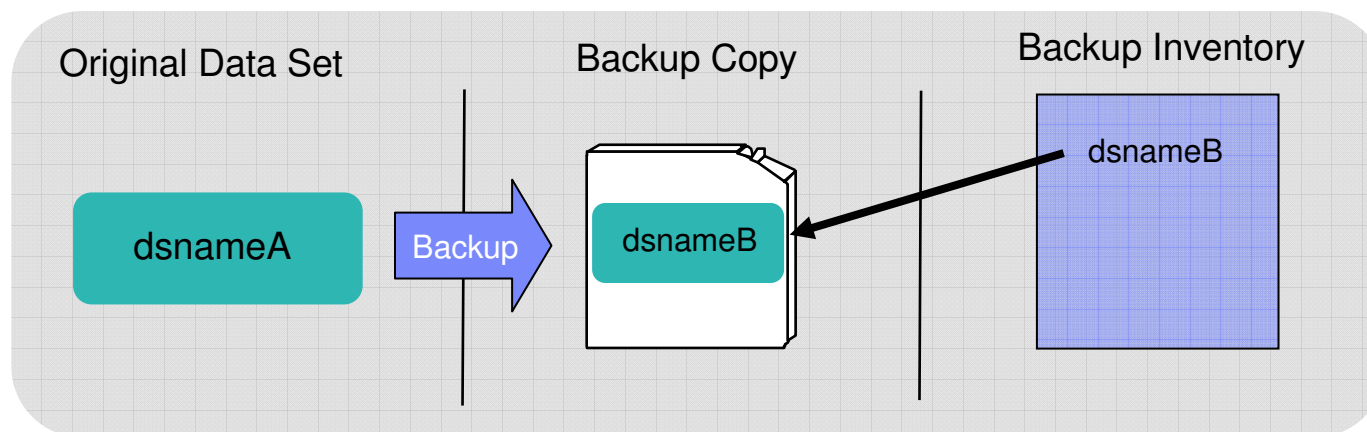
Virtual Concurrent Copy

- Concurrent Copy (CC)
 - Uses Control Unit Cache
 - Larger volumes cause constraint problems
- Virtual Concurrent Copy (SnapShot)
 - Data set is “snapped” to Working Space Data Set (WSDS)
 - After snap, backup is logically complete
 - After snap copy is moved to target device, backup is physically complete
- Virtual Concurrent Copy (FlashCopy) *New*
 - Supported on volumes that are CC and FlashCopy V2 capable
 - Data is Flashed using NOCOPY to WSDS
 - After Flash, backup is logically complete
 - After standard I/O is used to move the data to target device, backup is physically complete
 - Supported for Data Set Backup and Full-Volume Dump

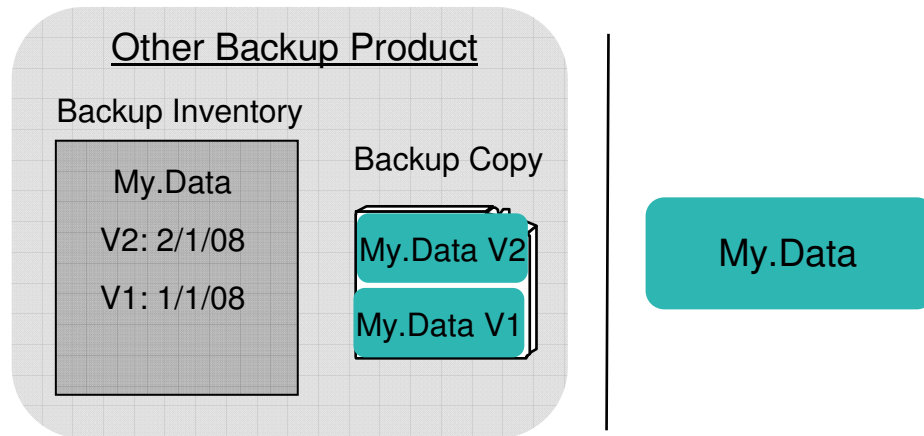
Data Set Backup New Name

- Specify a different name to store the backup copy in the DFSMSHsm inventory
 - Backup copy is created using the new name
 - DFSMSHsm inventory tracks backup copy using the new name
- Use to convert from another backup product to DFSMSHsm
 - Conversion done without impacting data availability

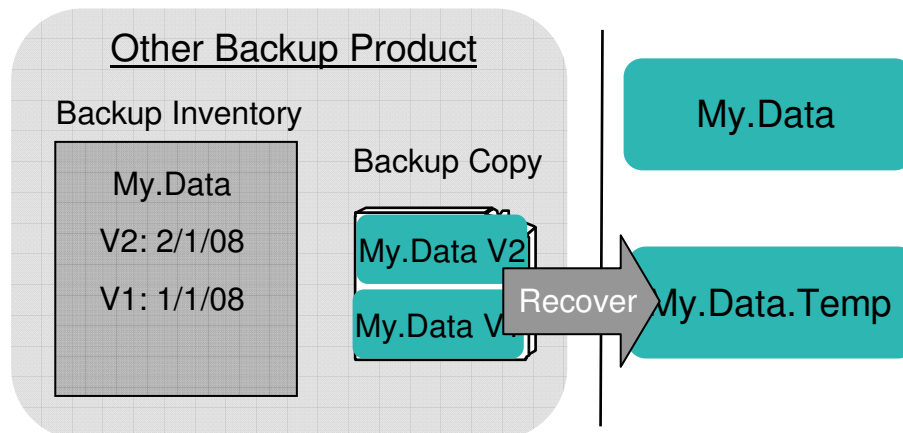
`BACKDS dsnameA NEWNAME(dsnameB)`



Converting to DFMSHsm Backup

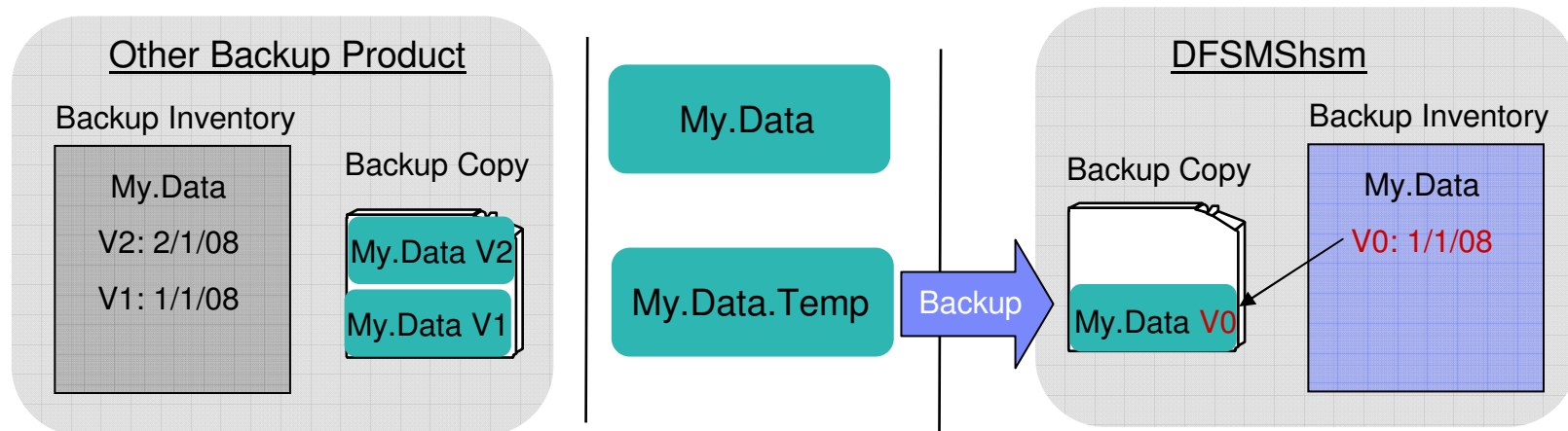


- Step 1: Recover an existing backup copy to a temporary name

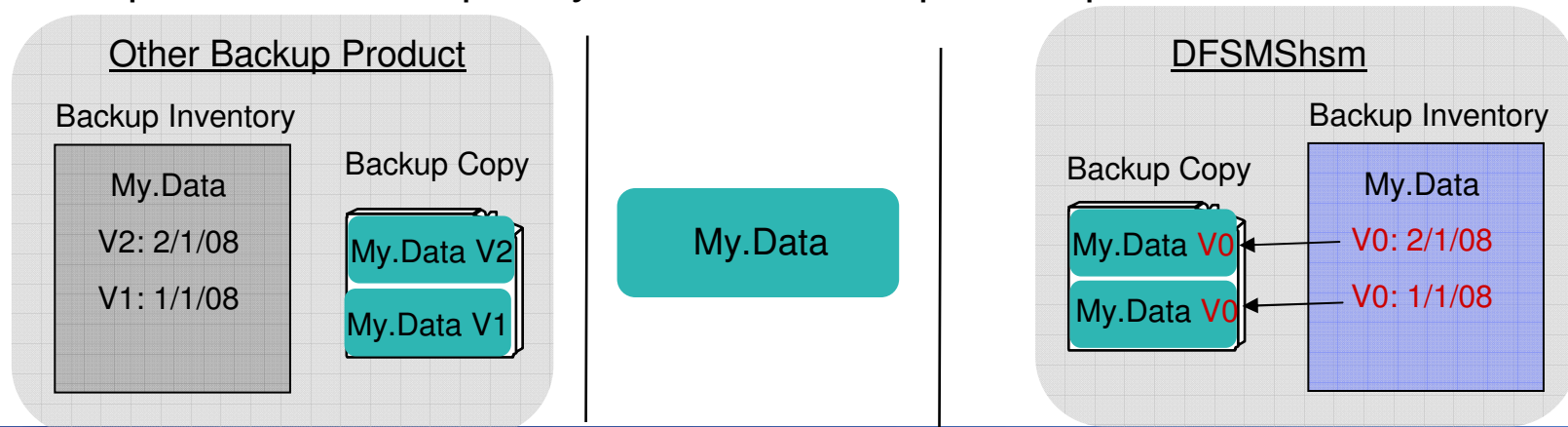


Converting to DFSMSHsm Backup

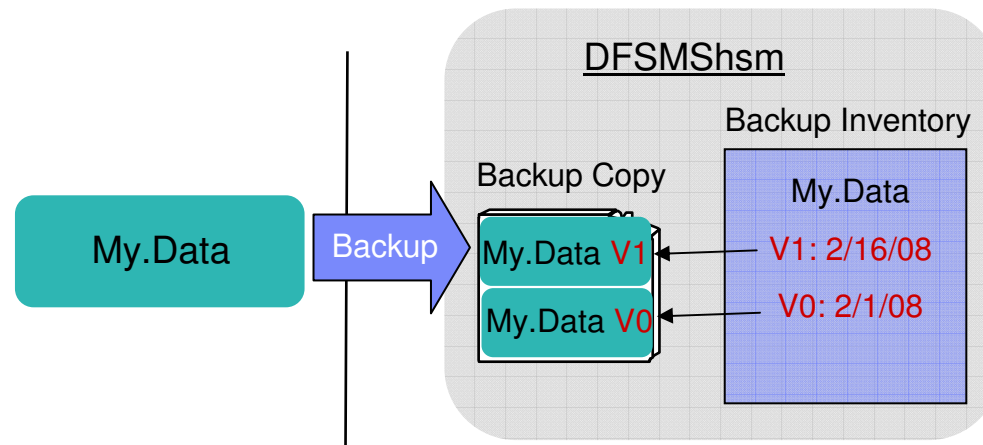
- Step 2: Backup with DFSMSHsm using NEWNAME and DATE



- Step 3: Delete Temporary data set and repeat steps 1 & 2



Converting to DFSMSHsm Backup



- Normal DFSMSHsm Backup copies can be created any time in the process
- New optional TIME parameter on both the data set backup and recover commands
- VERSION cannot be used to Recover a VER0 backup copy
- Recovery has no special processing

Small Requirements

- LIST TTOC SELECT(RESUMEAUDIT)
 - List tapes that were interrupted during AUDIT MEDIACONTROLS FIX and are eligible to have the Audit resume processing
- BD exit is called during individual data set backup
 - Previously, only invoked during Volume-level backup
- Use esoteric tape name in TCN record
 - Tape Copy Needed (TCN) record is created when a duplex copy fails
 - Previously, DFSMSHsm stored the generic tape name in the record, which could lead to allocation failures

DFSMSHsm Health Checks

Health Checks for DFSMSHsm

The objective of IBM Health Checker for z/OS (HC) is to identify potential problems before they impact availability or, in worst cases, cause outages.

- Checks are written by IBM, ISV's, and sometimes the customers themselves.
- Once a check is registered with HC, HC executes the check based on the interval defined for the check and/or when a check is requested to be run.
- After z/OS V1R8, checks can be LOCAL (execute in HC address space) or REMOTE (execute in a different address space).
- Check results can be viewed in a panel in SDSF.
- An exception is issued when a check finds a potential problem. Exceptions result in a message to the console. High severity exceptions result in a highlighted message to the console.

Health Checks for DFSMShsm

- **Three checks created to identify CDS backup problems**
 - HSM_CDSB_BACKUP_COPIES : Determines if HSM is configured to maintain a critical level of CDS backups. (Med by default)
 - HSM_CDSB_DASD_BACKUPS: Ensures all required DASD CDS backup data sets are in place (High by default)
 - HSM_CDSB_VALID_BACKUPS: Determines if the number of valid Control Data Set (CDS) backups has fallen below a critical level. (Med by default)
- Checks available for DFSMShsm V1R7.0 and above
 - [OA19262; Summer 2007](#)
- All 3 checks are LOCAL checks (run in HC address space)

Enabling the DFSMShsm Health Checks

- The first MAIN HSM host to start on a system will register the checks with z/OS Health Checker.
- The Health Checker address space does not have to be active when HSM registers the check. Checks will be registered but will not run until after the HC address space is started.
- A MAIN HSM host must be active for check to run to completion. If a check runs when a MAIN HSM host is not active, it will be disabled until the MAIN HSM host starts.

Successful Check Output

Display Filter View Print Options Help

```
-----  
SDSF OUTPUT DISPLAY HSM_CDSB_VALID_BACKUPS          LINE 0          COLUMNS 02- 133  
COMMAND INPUT ==>                                     SCROLL ==> PAGE
```

```
***** TOP OF DATA *****
```

```
CHECK(IBMHSM,HSM_CDSB_VALID_BACKUPS)
```

```
START TIME: 04/02/2007 13:41:20.065503
```

```
CHECK DATE: 20070316 CHECK SEVERITY: MEDIUM
```

```
CHECK PARM: CRITVAL(4)
```

```
Critical Value          : 4 (specified by owner)
```

```
BACKUPCOPIES specified  : 5
```

```
MCDS Valid Backups: 5
```

```
BCDS Valid Backups: 5
```

```
OCDS Valid Backups: 5
```

```
JRNL Valid Backups: 5
```

```
ARCHC0116I The check ran successfully and found no exceptions. The  
number of valid Control Data Set backup copies is equal to or greater  
than the critical value.
```

```
END TIME: 04/02/2007 13:41:20.122378 STATUS: SUCCESSFUL
```

```
***** BOTTOM OF DATA *****
```

Exception Check Output

Display Filter View Print Options Help

```
-----
SDSF OUTPUT DISPLAY HSM_CDSB_DASD_BACKUPS          LINE 0          COLUMNS 02- 133
COMMAND INPUT ==>                                SCROLL ==> PAGE
***** TOP OF DATA *****
CHECK(IBMHSM,HSM_CDSB_DASD_BACKUPS)
START TIME: 04/02/2007 10:48:49.673163
CHECK DATE: 20070316  CHECK SEVERITY: HIGH
```

BACKUPCOPIES specified : 5

MCDS Backups:

```
=====
Data Set Name                               Status
=====
DFHSM.MCDS.BACKUP.V0000001                *** NOT FOUND ***
DFHSM.MCDS.BACKUP.V0000002                *** NOT FOUND ***
DFHSM.MCDS.BACKUP.V0000003                *** NOT FOUND ***
```

...

* High Severity Exception *

ARCHC0111E One or more required Control Data Set backup copies not found.

Explanation: The check has found that DFSMSHsm backs up its CDS(s) to
DASD.

V1R9 Overview

DFSMSdss Cross Memory

In the GA version of V1R9, the default is to have DFSMSdss run in cross-memory mode for all functions *except* for Recall

- Intent was to increase the available storage in the DFSMSHsm address space to reduce occurrences of out-of-storage abends
- Other buffers were also moved above-the-line
- Some V1R9 customers have complained
 - RACF changes were not well documented
 - Don't want default to change from prior releases
 - Some have reported an increase in CPU utilization
- OA23236 opened to address these issues
 - Fully documents RACF implications
 - Changes the default to NOT invoke DFSMSdss in cross-memory mode
 - Documents the potential for an increase in CPU utilization

DFSMSdss Cross Memory

- A DFSMSdss Address Space is started for each function on each host
- Increases the available memory in DFSMSHsm
- The address space identifier for each nonFast Replication** function will be in the format of “ARCn`aaaa`”, where
 - `n` is the unique DFSMSHsm Host ID
 - `aaaa` represents the function, as follows:
 - “DUMP” – Dump
 - “REST” – Full-volume Recover
 - “MIGR” – Migration
 - “BKUP” – Backup
 - “RCVR” – Data Set Recover
 - “CDSB” – CDS Backup
 - Example: ARC1MIGR is the name of the DFSMSdss address space started for migration tasks on Host 1

** Fast Replication introduced other DFSMSdss address spaces in V1R8

Return Priority (RP) Exit

- Normal recall processing uses ACS routines. If a storage class is returned, DFSMSdfp selects the target volume.
- When the recall is for extent reduction, DFSMShsm does *not* invoke ACS but always targets the original volume.
- Requirement submitted for DFSMShsm to use ACS routines to determine the target volume during extent reduction :
- Enhancement:
 - Field added to the RP exit output parameter list.
 - If recall for extent reduction, DFSMShsm checks the new field in the output parameter list.
 - When set to 1, HSM will invoke the ACS storage class routine and modify the target volume as required.

FSR Enhancements

Requirements were submitted for the following FSR fields to be added to DFSMSHsm:

- Indicate in the tape recall FSR (type 5) when a recall caused a tape takeaway
- Record in the tape recall FSR (type 5) the number of times a migrated data set had been recycled before it was eventually recalled
- Add the recycle source volume volser to the recycle FSR (types 10 and 12)
- Provide the CPU time used for a partial release (type 18) and the expiration of a backup version (type 19)
- Record the number of tracks needed when an error occurred due to not enough ML1 space for migrate and freevol FSR (types 1 and 2)
- Update DFSMSHsm's ARCTOOLS sample REXX (FSRSTAT)

V1R8 Quick Overview

Alter Priority

- New command to alter priority of queued requests

**ALTERPRI (REQUEST(*reqnum*) | USERID(*userid*) |
DATASETNAME(*dsname*)) HIGH | LOW**

- Issue the **QUERY REQUEST** command to identify the *reqnum*, *userid*, or *dsname* of the request to be reprioritized
- **HIGH** is the default and moves the request to the front of the corresponding functional queue
- Specifying the **LOW** keyword moves the request to the back of the functional queue
- ✓ Use this new command to reprioritize queued requests on an exception basis
- The new function is intended to allow expediting a request which has already been queued, not to serve as the primary means to set priority values
 - The **RPEXT** can be used to assign priority values to recall, delete, and recover MWEs as they are put onto the queue

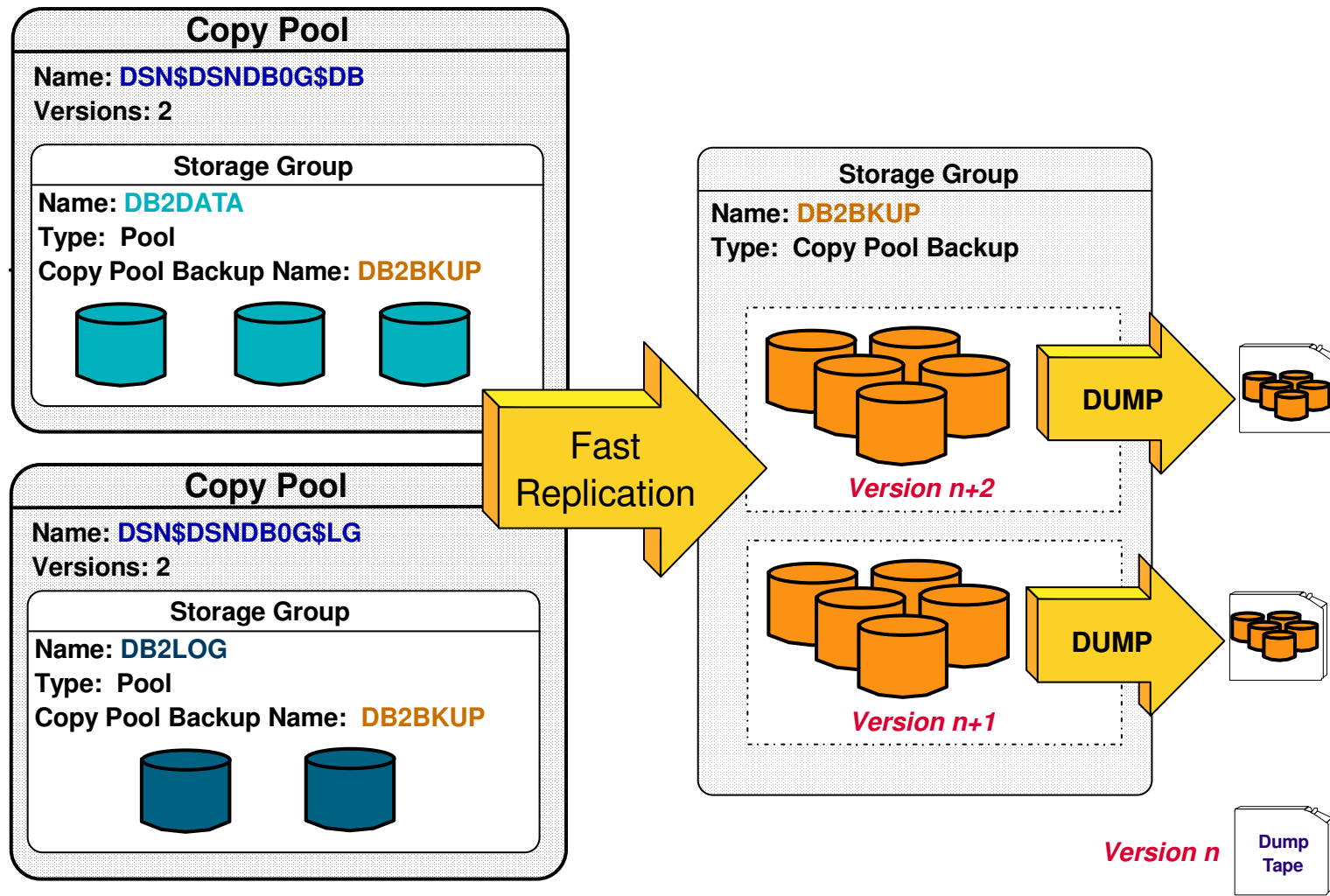
DFSMSHsm Fast Replication Overview

- DFSMSHsm invokes volume-level *fast replication* to create backup versions for sets of volumes
 - Backup hundreds of volumes in a matter of minutes
 - Developed jointly with DB2
 - System z Continuous Data Protection (zCDP) solution
- Generic term of “*Fast Replication*”
 - Multiple Point-in-Time copy functions supported by DFSMSdss
 - FlashCopy ® (Any vendor disk that supports FlashCopy interface)
 - SnapShot ®
- Recovery can be performed for the entire volume set, individual volumes or individual data sets from disk or tape

Fast Replication Enhancements

- Dump target volumes to tape
 - By command
 - Via Automatic Dump
- Additional types of FlashCopy
 - Incremental FlashCopy
 - NOCOPY
- Recover individual data sets from the copy pool backup
 - Either from disk or tape

DFSMSHsm Fast Replication Overview



Other V1R8 Enhancements

- **Recycle performance improvement**
 - SYNCHDEV performed at intervals instead of for every data set
 - No externals
- **Improved error handling on duplex tapes**
 - Optionally, both original and alternate migration tapes marked full when error occurs on alternate
 - New **SETSYS DUPLEX(MIGRATION(Y ERRORALTERNATE(CONTINUE | MARKFULL)))** option for migration
 - ★ *Guarantees that original and alternate are always identical*
 - No delay creating alternate copy

Other V1R8 Enhancements

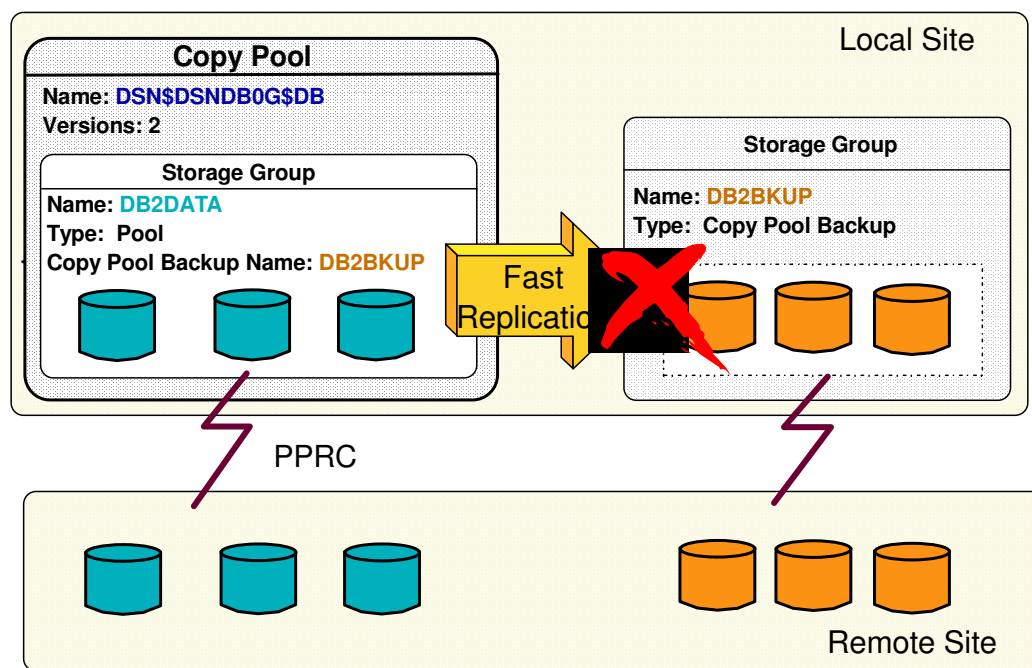
- **Migration Scratch queue for Non-VSAM data sets**
 - Improve performance of volume level migration activity when there are a large number of non-VSAM data
 - No external changes

- **Individual Data Set Restore for ABARS**
 - Allows restoring either a single data set or a group of data sets from an aggregate backup tape
 - No data set inventory maintained. Customer must determine which aggregate backup version contains the needed dataset

Development APARS

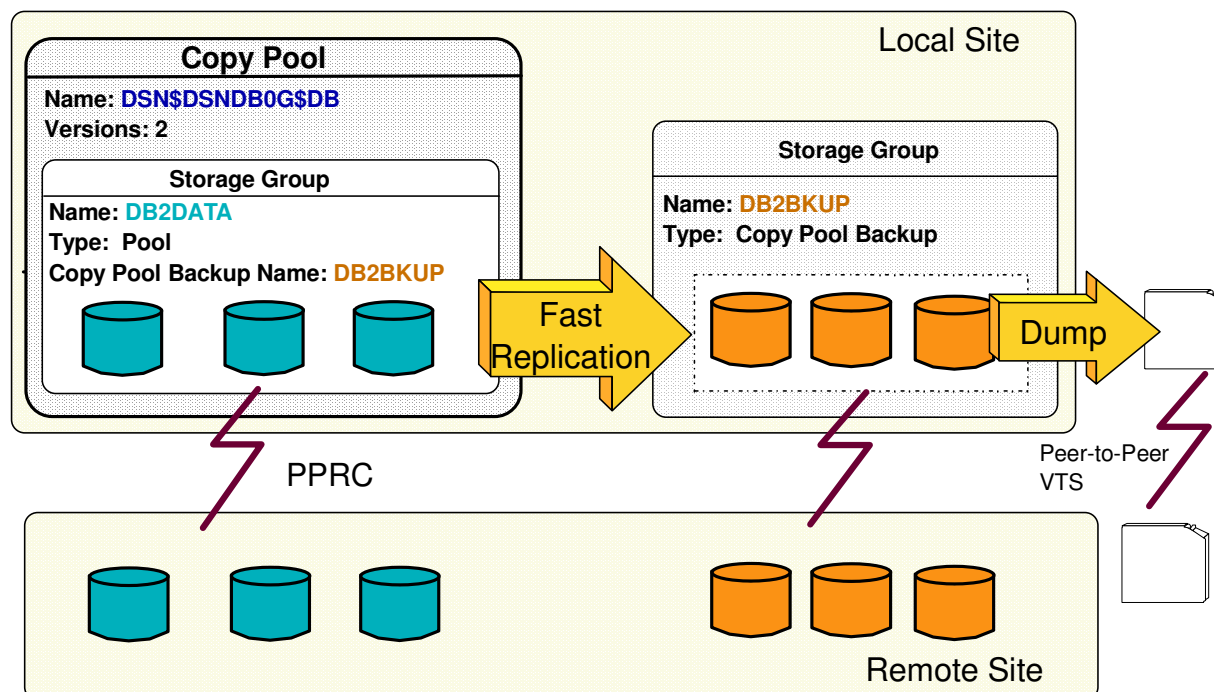
FlashCopy to PPRC Primary

- Without new support, DFSMSHsm does not support mirroring the copy pool target volumes
- This means that you have to break pairs before each FlashCopy and re-establishing pairs after FlashCopy



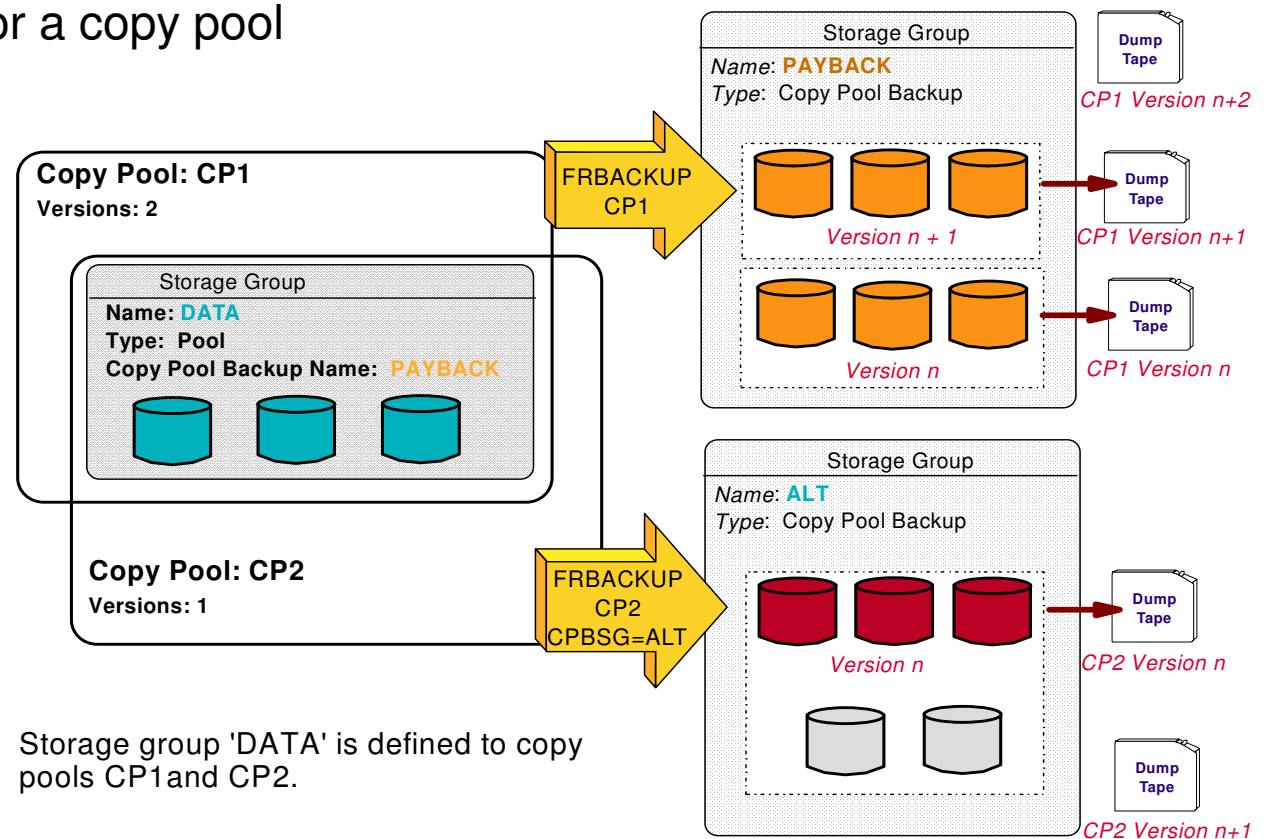
FlashCopy to PPRC Primary

- OA23849 – Enable FlashCopy to a PPRC Primary Volume
 - Enables easier implementation of backup volumes at remote site
 - Pair is put into a 'Duplex Pending' state until volumes re-synched
 - Enabled for Recovery, but you generally don't want production volumes to go in a 'Duplex Pending' state



Alternate Copy Pool Backup

- OA21666 - Alternate copy pool backup storage group
 - Enables a different copy pool backup storage group to be specified for a copy pool
 - Can be used to direct special copies to a specified set of backup volumes



Recall by FBID

- **OA24053** - Recall by File Block ID
 - When a large number of data sets are recalled concurrently from the same tape, recalling by FIFO causes excessive tape repositioning

SETSYS TAPEDATASETORDER(PRIORITY | FBID)

- FBID causes noWait-type requests to be processed before Wait-type requests

Summary

- Significant number of new and enhanced functions
 - Emphasis on addressing customer requirements
- Some critical new functions provided as “Development APARS”
- Expanded functionality of DFSMSHsm Fast Replication Support
- These items demonstrate IBM’s commitment to continual improvement of the DFSMSHsm product

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